

Lecture 9 — American and Bermudan options

Study notes: optimal stopping, the free boundary, the LCP, and pricing by tree / FD /
LSM / COS

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Abstract

These notes work through Lecture 9 in full. The early-exercise feature turns a plain expectation into an *optimal-stopping* problem with a *free boundary*: at every monitoring instant the holder takes the larger of the intrinsic payoff and the continuation value, and the set of states where exercise wins is bounded by an unknown curve $S^*(t)$. Two facts orient everything. (i) For a *non-dividend* underlying the American *call* is worth exactly the European call — early exercise is never optimal, by a clean no-arbitrage argument. (ii) The American *put* has no such equivalence: early exercise can be strictly optimal, the price solves a *linear complementarity problem* (LCP), and it must be priced numerically. We cover all four pricing routes on the slides — binomial tree, finite differences with the LCP (the projection / PSOR step), Longstaff–Schwartz regression Monte Carlo (LSM), and the COS/Bermudan backward recursion (Fang–Oosterlee) — and close with Richardson extrapolation from Bermudan to American. The COS section leans on the Lecture 6 machinery (the cosine coefficients A_k read off the characteristic function, and the χ_k, ψ_k building-block integrals); those derivations are not repeated, only invoked. See `notes/Lecture06-notes.pdf`.

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1 The early-exercise problem and the free boundary

An American option may be exercised at *any* time $t \in [0, T]$; a Bermudan option at a discrete set of monitoring dates $\zeta = \{t_1, \dots, t_N\}$ with $t_N = T$. The continuous American case is the limit of the Bermudan as the monitoring set fills $[0, T]$; *every numerical method below in fact prices a Bermudan*, and the American value is obtained by letting $N \rightarrow \infty$ (or by Richardson extrapolation, §9).

At each admissible instant the holder solves a binary decision: *exercise now* for the intrinsic payoff, or *hold* and keep the option alive. Writing Λ for the intrinsic payoff (a call pays $(S_t - K)^+$, a put $(K - S_t)^+$), the value is the larger of the two,

$$v(S, t) = \max \{ \Lambda(S, t), \text{continuation value at } (S, t) \}. \quad (1)$$

The continuation value is the discounted risk-neutral expectation of the option value an instant later. This is a *dynamic-programming / optimal-stopping* structure: the price is

$$v(S_0, 0) = \sup_{\tau \in \mathcal{T}_{[0, T]}} \mathbb{E}^Q [e^{-r\tau} \Lambda(S_\tau, \tau) \mid S_0], \quad (2)$$

the supremum over all stopping times τ adapted to the price filtration. The hard part is that the optimal τ^* is *not known in advance*; it is itself an output of the problem.

The free boundary. The state space (S, t) splits into two regions:

- the **continuation region** $\mathcal{C}$, where it is optimal to hold and $v(S, t) > \Lambda(S, t)$ strictly; here v satisfies the Black–Scholes PDE with equality;
- the **stopping (exercise) region** $\mathcal{S}$, where it is optimal to exercise and $v(S, t) = \Lambda(S, t)$.

The interface between them is the *optimal exercise boundary* $S^*(t)$. It is a “free” boundary because its location is unknown a priori and must be solved for together with v — in contrast to a fixed Dirichlet/Neumann boundary. For the American put (slide 10): if S is large it is never worth exercising (you would throw away the remaining optionality and the interest on K), so it is optimal to hold; as $S \rightarrow 0$ the payoff $K - S$ approaches its maximum and is best taken immediately. Hence there is a well-defined $S^*(t)$ with

$$S_t < S^*(t) \Rightarrow \text{exercise, } v = K - S_t, \quad S_t > S^*(t) \Rightarrow \text{hold, } v > K - S_t. \quad (3)$$

Across the boundary v and $\partial_S v$ match continuously (the *smooth-pasting* conditions $v(S^*, t) = K - S^*$, $\partial_S v(S^*, t) = -1$); the slides state the existence of $S^*(t)$ but do not derive smooth pasting. [Marked as not on the slides — standard result, stated for completeness.]

2 American call on a non-dividend stock equals the European call

Remark 1 (Claim). *If the underlying pays no dividends, the American call has the same value as the European call: it is never optimal to exercise early.*

Slide argument (no-arbitrage, dominating strategy). Suppose at some $t < T$ the holder considers exercising the call. This is only conceivably worthwhile if $S_t > K$, giving payoff $S_t - K$. Compare against the following *alternative strategy* that keeps the option alive:

1. At t , **short** the underlying, receiving S_t in cash, and **keep** the (unexercised) call.
2. At maturity T , close the short by buying one unit of the asset, choosing the cheaper of:
 - exercising the call at maturity, paying K (used when $S_T > K$); or
 - buying in the open market at S_T (used when $S_T \leq K$).

In either case the cash paid at T to acquire the asset is $\min(K, S_T) \leq K$.

Compare the two cash flows. Early exercise gives $S_t - K$ at time t . The alternative gives S_t at time t and pays out at most K at the later time T . Even ignoring the time value of money, S_t now and $\leq K$ later already weakly dominates $S_t - K$ now (the alternative keeps the full S_t and the cash payment K is deferred and possibly smaller). Accounting for interest the dominance is strict: holding K in the bank from t to T earns interest, so the present value of the deferred $\leq K$ outflow is strictly less than K . Therefore early exercise is dominated, and since this holds at every $t < T$, the early-exercise feature is worthless: the American call equals the European call.

Cleaner version (lower-bound argument). An equivalent and very quotable proof: for a *non-dividend* stock the European call satisfies, by put-call parity and $P^{\text{eur}} \geq 0$,

$$C^{\text{eur}}(S, t) = S - Ke^{-r(T-t)} + P^{\text{eur}}(S, t) \geq S - Ke^{-r(T-t)} > S - K \quad (4)$$

whenever $r > 0$ and $t < T$. The right-hand side $S - K$ is the intrinsic value (the payoff from exercising now). Thus the live European call is worth strictly *more* than its immediate-exercise value: the holder would never trade the call for the intrinsic, so the early-exercise right is never used and adds nothing, $C^{\text{am}} = C^{\text{eur}}$. The economic content of $-Ke^{-r(T-t)}$ versus $-K$ is exactly the interest earned by deferring the strike payment; with dividends, the asset can “leak” value before T and the inequality (4) can fail, which is why dividend-paying calls *can* be exercised early.

Why the put is different. For the put, parity gives $P^{\text{eur}} = Ke^{-r(T-t)} - S + C^{\text{eur}}$, and $Ke^{-r(T-t)} < K$, so the live European put can be worth *less* than its intrinsic value $K - S$. When that happens, exercising immediately and banking $K - S$ (to earn interest on the full K) beats holding. So early exercise can be strictly optimal: there is no European–American equivalence for the put, and we must price it numerically (§3 onward). Intuitively, the put holder *wants* the strike cash K early to earn interest, whereas the call holder *wants* to defer paying K — the asymmetry is entirely in the sign of the strike cash flow.

3 Black–Scholes for the American put: the LCP / obstacle formulation

We write $P_{\text{Am}}(S, t)$ for the American put value. Two inequalities pin it down, and at every point *one of them holds with equality*.

(I) Value dominates payoff. Trivially, by no-arbitrage,

$$P_{\text{Am}}(S, t) \geq \max(K - S, 0). \quad (5)$$

If $P_{\text{Am}} < \max(K - S, 0)$ one could buy the option and exercise it immediately for an instant riskless profit.

(II) The PDE becomes an inequality. Form the delta-hedged portfolio $\Pi = \Delta S + D$ with $\Delta = \partial P_{\text{Am}} / \partial S$, exactly as in the European BS derivation. For a European option the hedged position is riskless and earns the risk-free rate: $\Delta(P - \Pi) = r(P - \Pi)\Delta t$. For the American put this becomes an *inequality*, because delta-hedging does not remove the uncertainty contributed by the holder's early-exercise right. The slides argue both directions by arbitrage:

- If $\Delta(P_{\text{Am}} - \Pi) > r(P_{\text{Am}} - \Pi)\Delta t$, the hedged option out-earns cash. An arbitrageur borrows at r , buys $(P_{\text{Am}} - \Pi)$ (long the put, short Π), thereby *owning the exercise right*, and locks in the surplus. This is a genuine arbitrage and is competed away — so this case cannot persist.
- If $\Delta(P_{\text{Am}} - \Pi) < r(P_{\text{Am}} - \Pi)\Delta t$, the hedged option under-earns cash. To arbitrage it one would *short* $(P_{\text{Am}} - \Pi)$ and deposit cash — but shorting the put hands the exercise decision to the *counterparty*, who may exercise during $[t, t + \Delta t]$. There is no longer a guaranteed riskless profit, so this case is *not* ruled out.

Only the second case survives, giving the one-sided relation $\Delta(P_{\text{Am}} - \Pi) \leq r(P_{\text{Am}} - \Pi)\Delta t$. Expanding $\Delta(P_{\text{Am}} - \Pi)$ with Itô and cancelling the Δt yields the **partial differential inequality**

$$\mathcal{L} P_{\text{Am}} := \frac{\partial P_{\text{Am}}}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 P_{\text{Am}}}{\partial S^2} + rS \frac{\partial P_{\text{Am}}}{\partial S} - rP_{\text{Am}} \leq 0. \quad (6)$$

(The slides write $\partial P / \partial S S$ for the drift term; under the risk-neutral measure the drift coefficient is r , giving $rS \partial_S P$.)

The linear complementarity problem. Combining (5) and (6), at every (S, t) *at least one is an equality*, and they cannot both be strict:

$$\boxed{(P_{\text{Am}} - \Lambda) \geq 0, \quad \mathcal{L} P_{\text{Am}} \leq 0, \quad (P_{\text{Am}} - \Lambda) \cdot (\mathcal{L} P_{\text{Am}}) = 0,} \quad (7)$$

with $\Lambda = \max(K - S, 0)$. This is the *linear complementarity problem* (equivalently an *obstacle problem*: P_{Am} is the smallest supersolution of the BS operator lying above the obstacle Λ). Reading the complementarity:

- in the continuation region, $P_{\text{Am}} > \Lambda$ so $\mathcal{L} P_{\text{Am}} = 0$ — the option satisfies the BS PDE exactly;
- in the stopping region, $P_{\text{Am}} = \Lambda$ so $\mathcal{L} P_{\text{Am}} \leq 0$ — the PDE is violated (the payoff is not a BS solution), and exercise is optimal.

The boundary where the active constraint switches is precisely the free boundary $S^*(t)$ of §1.

Auxiliary conditions. Terminal condition $P_{\text{Am}}(S, T) = \max(K - S, 0)$. Boundary conditions $P_{\text{Am}}(0, t) = K$ (deep in the money the put is worth K , exercised now) and $\lim_{S \rightarrow \infty} P_{\text{Am}}(S, t) = 0$ (far out of the money it is worthless), for $0 \leq t \leq T$.

4 Bermudan formulation in log-space (the common backbone)

All numerical methods are cleanest in the log variable. Let t_0 be now, $\zeta = \{t_1, \dots, t_N\}$ the exercise dates, $\Delta t = t_n - t_{n-1}$, $t_N = T$, and

$$X(t_n) \equiv \ln \frac{S(t_n)}{K}. \quad (8)$$

The payoff is $\Lambda(x, t) = \max(\alpha K(e^x - 1), 0)$ with $\alpha = +1$ for a call, $\alpha = -1$ for a put. The backward recursion, for $n = N, N - 1, \dots, 2$, reads

$$c(x, t_{n-1}) = e^{-r\Delta t} \mathbb{E}[v(X(t_n), t_n) \mid X(t_{n-1}) = x], \quad (9)$$

$$v(x, t_{n-1}) = \max(\Lambda(x, t_{n-1}), c(x, t_{n-1})), \quad (10)$$

terminating at

$$v(x, t_0) = e^{-r\Delta t} \mathbb{E}[v(X(t_1), t_1) \mid X(t_0) = x], \quad v(x, T) = \Lambda(x, T). \quad (11)$$

Here c is the *continuation value* (the conditional expectation (9)) and $v = \max(\Lambda, c)$ implements (1). The whole numerical challenge is evaluating the conditional expectation (9); the four methods below differ *only* in how they do that.

5 Binomial tree (the cleanest backward induction)

The recombining CRR tree carries this structure exactly. Build the tree to expiry $T = N\Delta t$ with up/down factors $u, d = 1/u$ and risk-neutral probability $p = (e^{r\Delta t} - d)/(u - d)$.

Terminal layer. At expiry the value is the payoff:

$$V_m^N = \max(K - S_m^N, 0), \quad 0 \leq m \leq M \text{ (put)}. \quad (12)$$

Backward step (the early-exercise max). For $n = N - 1$ down to 0, at each node take the larger of intrinsic and discounted expected continuation:

$$V_m^n = \max\left(\underbrace{K - S_m^n}_{\text{intrinsic}}, \underbrace{e^{-r\Delta t} [p V_{m+1}^{n+1} + (1-p) V_m^{n+1}]}_{\text{continuation} = c}\right), \quad 0 \leq m \leq n. \quad (13)$$

The conditional expectation (9) is here the two-point binomial average; the max enforces (10). The price is V_0^0 , read off in the final step. For a European option one would drop the intrinsic from the max and just discount the expectation — the single line (13) is the entire difference between European and American on a tree, and it is the canonical illustration of $\max(\text{continuation}, \text{intrinsic})$. A node where the intrinsic wins is in the stopping region; the boundary between exercised and held nodes traces the discrete analogue of $S^*(t)$.

6 Finite differences with the LCP (projection / PSOR)

Grid. Time lattice $t_n = n\Delta t$, $n = 0, \dots, N$; space lattice in $x = \ln(S/K)$, $x_m = m\Delta x$, $m = -M, \dots, M$. Let $V_m^n \approx v(x_m, t_n)$.

Terminal and boundary data.

$$V_m^N = \max(K - S_m^N, 0), \quad V_{-M}^n = K, \quad V_M^n = 0 \quad (n = 1, \dots, N), \quad (14)$$

matching the put boundary conditions of §3.

Discretised operator. Discretise the BS PDE; the slides use a backward-time, centred-space (BTCS, i.e. implicit) stencil, abbreviated by a stencil function $g(\cdot, \cdot, \cdot)$ acting on the three neighbouring nodes of the next time level:

$$g(V_{m+1}^{n+1}, V_m^{n+1}, V_{m-1}^{n+1}) \text{ (the implicit BS update at node } m). \quad (15)$$

Implicit time-stepping is preferred here for unconditional stability and because the LCP projection (below) composes cleanly with it.

The early-exercise projection. At each interior node, after (or within) the implicit solve, project the value back up onto the payoff:

$$V_m^n = \max\left(K - S_m^n, g(V_{m+1}^{n+1}, V_m^{n+1}, V_{m-1}^{n+1})\right), \quad m = -M + 1, \dots, M - 1. \quad (16)$$

This is the discrete form of the LCP (7): the implicit operator enforces $\mathcal{L}V \leq 0$, the max enforces $V \geq \Lambda$, and at each node exactly one is active — the complementarity. The price is V_0 at x_0 in the final backward step.

Why a naive “solve-then-clip” is not enough; PSOR. For an *explicit* scheme the right-hand side is known, so (16) is a literal solve-then-clip. For an *implicit* scheme the update at node m couples to its neighbours *at the same time level* through a tridiagonal system, and the constraint $V \geq \Lambda$ must hold simultaneously with the linear solve. The standard remedy is *Projected SOR (PSOR)*: run successive over-relaxation on the tridiagonal system, but after each Gauss–Seidel/SOR sweep update of node m , immediately project,

$$V_m^{n,(\text{new})} = \max\left(\Lambda_m, V_m^{n,(\text{SOR})}\right), \quad (17)$$

and iterate the sweeps to convergence. Projecting *inside* the iteration (not once at the end) is what makes PSOR converge to the true LCP solution rather than to the unconstrained solve clipped afterwards. [The slides give the projected update (16) explicitly but do not name PSOR or the implicit coupling subtlety; added here because it is the natural deep-dive and Fang may push on “why can’t you just solve the linear system and then take the max”.]

7 Longstaff–Schwartz regression Monte Carlo (LSM)

7.1 Why plain Monte Carlo fails

The Bellman recursion (9)–(10) needs the conditional expectation $c(x, t_{n-1})$ at every node of every path. A naive MC would have to launch a *nested* simulation from each node to estimate it — combinatorially infeasible. The Longstaff–Schwartz (2001) insight: *estimate the continuation value by cross-sectional least-squares regression* of realised discounted future cash flows onto basis functions of the current state, using the *same* set of paths. One outer simulation, no nesting.

7.2 The regression mechanics

Simulate P risk-neutral paths $\{S_{t_n}^p\}$. Work *backwards*. At the final date $t_N = T$ the cash flow is the terminal payoff. At each earlier exercise date t_n :

1. **Select in-the-money paths.** Only ITM paths face a real exercise decision (an OTM path has zero intrinsic, so it would never exercise); restricting the regression to ITM paths is both cheaper and more accurate (it fits the conditional expectation only where the decision matters).
2. **Build the regression data.** For each selected path p , let $X_p = S_{t_n}^p$ (the current state) and Y_p = the realised *discounted* cash flow obtained from that path by following the *already-determined* optimal policy from t_{n+1} onward (discounted back to t_n). Since the option is exercised at most once, Y_p is the discount of the single future cash flow on that path.
3. **Regress.** Fit Y on a small basis in X — the slides use a constant, X , X^2 (low-order polynomials; Laguerre polynomials in the original paper):

$$\mathbb{E}[Y | X] \approx \sum_{j=0}^J a_j L_j(X), \quad \hat{a} = \arg \min_a \sum_{p \in \text{ITM}} \left(Y_p - \sum_j a_j L_j(X_p)\right)^2. \quad (18)$$

The fitted function $\hat{c}(x) = \sum_j \hat{a}_j L_j(x)$ is the estimated continuation value.

4. **Exercise decision per path.** For each ITM path compare intrinsic against fitted continuation: exercise at t_n iff $\Lambda(X_p) > \hat{c}(X_p)$. Where exercise wins, set that path's cash flow to the intrinsic at t_n and *zero out* any later cash flows (the option is now spent); where it loses, keep the future cash flow.

Roll back to t_{n-1} and repeat. After reaching t_1 each path has a single stopping time. The price is the average over paths of the cash flow discounted to t_0 ,

$$\hat{v}_0 = \frac{1}{P} \sum_{p=1}^P e^{-r\tau_p} \Lambda(S_{\tau_p}^p, \tau_p). \quad (19)$$

7.3 The worked 8-path example (slides 13–19)

American put, $K = 1.10$, $r = 6\%$, three dates $t = 1, 2, 3$; one-period discount $e^{-0.06} = 0.94176$.

- **$t = 2$ regression.** Five ITM paths; $Y =$ discounted ($\times 0.94176$) $t = 3$ payoff, $X = S_{t=2}$. The least-squares fit is $\mathbb{E}[Y \mid X] = -1.070 + 2.983X - 1.813X^2$. Comparing intrinsic vs. fitted continuation, exercise is optimal at $t = 2$ on paths 4, 6, 7.
- **$t = 1$ regression.** Five ITM paths; Y now the discounted value of the *subsequent* cash flow (which by the $t = 2$ decision occurs at either $t = 2$ or $t = 3$, discounted back to $t = 1$). The fit is $\mathbb{E}[Y \mid X] = 2.038 - 3.335X + 1.356X^2$, and exercise is optimal at $t = 1$ on paths 4, 6, 7, 8.
- **Stopping rule and price.** The exercise-indicator matrix has a single 1 per exercised path (at the *first* optimal date). Discount each path's realised cash flow to time 0 and average: $\hat{v}_{\text{Am}} = 0.1144$, versus the European put 0.0564 from discounting only the $t = 3$ payoffs — the American put is worth roughly twice the European, the entire value of the early-exercise right.

7.4 Mathematical statement and a caution

No-arbitrage gives the continuation value as a conditional expectation of remaining discounted cash flows under Q ,

$$F(\omega; t_k) = \mathbb{E}^Q \left[\sum_{j=k+1}^N \exp\left(-\int_{t_k}^{t_j} r(\omega, s) ds\right) C(\omega, t_j; t_k, T) \middle| \mathcal{F}_{t_k} \right], \quad (20)$$

and LSM replaces this by the least-squares projection (18) onto finitely many $\mathcal{F}_{t_k}$ -measurable basis functions, backwards from t_{N-1} to t_1 .

Remark 2 (Why LSM is low-biased, and the in-sample subtlety). *The estimated policy is suboptimal (a finite basis cannot represent the true continuation surface exactly), so (19) is in principle a lower bound on the true price — a suboptimal stopping rule can only lose value. There is a competing upward bias if the price is read off the same cash flows used to fit the regression (in-sample foresight). The clean practice is to use the regression only for the exercise decision and then evaluate the realised cash flows along each path's stopping time (as in (19)); for a strictly out-of-sample estimate one re-simulates fresh paths and applies the frozen policy.* [Not on the slides; standard and a likely follow-up.]

8 The COS / Bermudan early-exercise method (Fang–Oosterlee)

This is the priority section. COS evaluates the conditional expectation (9) with no nested simulation and no quadrature, because it uses the *density* directly via the characteristic function. It reuses the entire Lecture 6 apparatus; see `notes/Lecture06-notes.pdf` for the derivations of the cosine coefficients and the χ_k, ψ_k integrals, which are only invoked here.

8.1 From the recursion to a cosine recursion

Write the continuation value (9) as an integral against the transition density $f(y \mid x)$ of the log-state:

$$c(x, t_{n-1}) = e^{-r\Delta t} \int_{-\infty}^{\infty} v(y, t_n) f(y \mid x) dy. \quad (21)$$

This is *exactly* the European COS integral of Lecture 6, but with the European payoff replaced by the *next-step option value* $v(\cdot, t_n)$. Approximate it by the COS quadrature (insert the cosine expansion of f , whose coefficients are read off the characteristic function φ):

$$c(x, t_{n-1}) \approx \hat{c}(x, t_{n-1}) \equiv e^{-r\Delta t} \sum_{k=0}^{K-1'} \Re \left\{ \varphi\left(\frac{k\pi}{b-a}\right) e^{ik\pi \frac{x-a}{b-a}} \right\} V_k(t_n), \quad (22)$$

where the prime halves the $k = 0$ term and $\varphi(\cdot)$ is the characteristic function of the increment $X(t_n) - X(t_{n-1})$ over one step Δt (for Lévy/BS models the conditional density is shift-invariant, so a single set of coefficients serves every x). The objects $V_k(t_n)$ are the *cosine coefficients of the option value at t_n* :

$$V_k(t_n) = \frac{2}{b-a} \int_a^b v(y, t_n) \cos\left(k\pi \frac{y-a}{b-a}\right) dy. \quad (23)$$

At maturity these are just the payoff coefficients of Lecture 6; for $n < N$ they are built from the previous step. This is the key structural fact: (22)–(23) is a *backward recursion on the coefficient vector* $\{V_k(t_n)\}_k$, not on a grid of prices.

8.2 Splitting at the early-exercise point

Because $v(y, t_n) = \max(\Lambda(y, t_n), c(y, t_n))$, the integral (23) splits at the *early-exercise point* x_n^* — the log-state at t_n where continuation equals intrinsic,

$$c(x_n^*, t_n) = \Lambda(x_n^*, t_n). \quad (24)$$

This x_n^* is the discrete-time avatar of the free boundary $S^*(t)$ of §1. On one side of x_n^* the option value equals the payoff Λ ; on the other side it equals the continuation c . Hence, with the building-block coefficient integrals

$$G_k(x_1, x_2) \equiv \frac{2}{b-a} \int_{x_1}^{x_2} \Lambda(y, t_n) \cos\left(k\pi \frac{y-a}{b-a}\right) dy, \quad (25)$$

$$C_k(x_1, x_2, t_n) \equiv \frac{2}{b-a} \int_{x_1}^{x_2} c(y, t_n) \cos\left(k\pi \frac{y-a}{b-a}\right) dy, \quad (26)$$

the coefficients split as

$$V_k(t_n) = \begin{cases} C_k(a, x_n^*, t_n) + G_k(x_n^*, b), & \text{call } (\alpha = +1), \\ G_k(a, x_n^*, t_n) + C_k(x_n^*, b, t_n), & \text{put } (\alpha = -1). \end{cases} \quad (27)$$

The geometry: a *put* is exercised for small y (low asset price), so Λ is active on $[a, x_n^*]$ and continuation on $[x_n^*, b]$ — giving $G_k(a, x_n^*) + C_k(x_n^*, b, t_n)$. A *call* is the mirror image. The G_k pieces are elementary: $\Lambda(y) = \alpha K(e^y - 1)$ on its active region, so G_k is exactly a combination of the Lecture-6 closed forms χ_k (the $e^y \cos$ integral) and ψ_k (the $\cos$ integral) — **the same** χ_k, ψ_k as in Lecture06-notes.pdf, evaluated on the sub-interval cut at x_n^* .

8.3 The C_k piece: where the previous step's coefficients re-enter

The non-trivial piece is C_k , the cosine coefficient of the *continuation function* $c(\cdot, t_n)$ over the hold region. Substitute the COS form $\hat{c}$ from (22) (with $t_{n-1} \rightarrow t_n$, i.e. written in terms of the *already-known* $V_\ell(t_{n+1})$) into the defining integral (26). Then C_k becomes a sum over ℓ of integrals of $\cos(k \cdots) \cos(\ell \cdots)$ and $\cos(k \cdots) \sin(\ell \cdots)$ over $[x_1, x_2]$ — all elementary, all available in closed form. Collecting them, C_k is a matrix–vector product acting on $\{V_\ell(t_{n+1})\}$:

$$C_k(x_1, x_2, t_n) = \frac{e^{-r\Delta t}}{\pi} \Im \left\{ \sum_{\ell=0}^{K-1} {}'\varphi\left(\frac{\ell\pi}{b-a}\right) V_\ell(t_{n+1}) [\mathcal{M}_{k,\ell}(x_1, x_2)] \right\}, \quad (28)$$

where $\mathcal{M}_{k,\ell}$ is the (analytically known) integral of the product of trig factors. *The crucial efficiency point:* the kernel matrix $\mathcal{M}$ has a Hankel-plus-Toeplitz structure, so the matrix–vector product (28) is computed by **FFT** in $O(K \log_2 K)$ rather than $O(K^2)$. [The slides state the χ/ψ -type integrals, the FFT acceleration, and the complexity but do not print the explicit Toeplitz/Hankel split of $\mathcal{M}_{k,\ell}$; I have summarised it as (28) — the load-bearing facts (closed-form trig integrals, FFT, $O(K \log K)$) are on the slides, the matrix labels are from Fang–Oosterlee (2009).]

8.4 The algorithm and its cost

1. **Initialise** at $t_N = T$ with the pure payoff coefficients: $V_k(t_N) = G_k(0, b)$ for a call, $G_k(a, 0)$ for a put (the payoff is active on the whole ITM half-range; these are the European V_k of Lecture 6).
2. **Backward loop**, $n = N - 1, \dots, 1$:
 - (a) Find the *early-exercise point* x_n^* by solving (24), $\hat{c}(x_n^*, t_n) = \Lambda(x_n^*, t_n)$, with **Newton's method** (one scalar root-find; the derivative of $\hat{c}$ is available analytically by differentiating the cosine sum).
 - (b) Form $V_k(t_n)$ via the split (27): the G_k pieces in closed form (χ_k, ψ_k) , the C_k piece by the FFT-accelerated product (28) acting on $\{V_\ell(t_{n+1})\}$.
3. **Final step:** evaluate $\hat{v}(x_0, t_0) = e^{-r\Delta t} \sum_k {}'\Re\{\varphi(\frac{k\pi}{b-a}) e^{ik\pi(x_0-a)/(b-a)}\} V_k(t_1)$ — one European-style COS sum against the t_1 coefficients.

Complexity $O(N K \log_2 K)$: N time steps, each an FFT of length K . The slide-24 picture: under both BS and CGMY, COS reaches $\sim 10^{-9}$ error in ~ 10 – 15 ms, sweeping past the CONV (FFT-based) method, which plateaus near 10^{-4} – 10^{-5} . The exponential convergence of European COS survives into the Bermudan recursion because each step's integrand inherits the smoothness of the density.

Remark 3 (Range caveat carries over — the Assignment-2 trap). *The truncation range is still set from the cumulants of X_T , $[a, b] = [c_1 - L\sqrt{c_2 + \sqrt{c_4}}, c_1 + L\sqrt{c_2 + \sqrt{c_4}}]$ (Lecture06-notes.pdf, §range), but now it must contain the support over the whole backward recursion, and the Newton search for x_n^* must land inside $[a, b]$. At short maturities (or short Δt per step) $c_2 = \sigma^2 \Delta t \rightarrow 0$, the box collapses toward c_1 , and (Gaussian: $c_4 = 0$) there is no $\sqrt{c_4}$ floor — the same failure mode as my Assignment-2 bug. Adding cosine terms K does not cure a range error; the fix is a wider L or a minimum half-width floor. This is exactly the kind of point Fang presses on.*

9 Richardson extrapolation: Bermudan $\rightarrow$ American

Every method above prices a *Bermudan* (finite monitoring set). The American value is the $N \rightarrow \infty$ limit, which converges but slowly. Let $v(M)$ be the Bermudan price with M exercise dates.

Rather than push M very large, take a handful of Bermudan prices at $M = 2^d, 2^{d+1}, 2^{d+2}, 2^{d+3}$ and combine them with the 4-point Richardson scheme

$$v_{\text{Am}}(d) = \frac{1}{21} \left(64 v(2^{d+3}) - 56 v(2^{d+2}) + 14 v(2^{d+1}) - v(2^d) \right), \quad (29)$$

which cancels leading error terms in the monitoring spacing and accelerates convergence to the continuous American limit. This is the standard, cheap way to get an American price from a small number of fast Bermudan COS computations.

10 One-paragraph reconstruction (to be able to say out loud)

An American option is an optimal-stopping problem: at each instant the holder takes $\max(\text{intrinsic}, \text{continuation})$ and the states where exercise wins are bounded by a free boundary $S^*(t)$. For a non-dividend call early exercise is never optimal — the live call $C^{\text{eur}} = S - K e^{-r(T-t)} + P^{\text{eur}} > S - K$ already beats its intrinsic, because deferring the strike payment earns interest — so American = European. The put is the opposite ($K e^{-r(T-t)} < K$ makes banking the strike early attractive), so it solves the linear complementarity problem $\{v - \Lambda \geq 0, \mathcal{L}v \leq 0, (v - \Lambda)\mathcal{L}v = 0\}$ and must be priced numerically. All methods run the same backward Bellman recursion $c = e^{-r\Delta t} \mathbb{E}[v(\cdot)]$, $v = \max(\Lambda, c)$, differing only in how they get the conditional expectation: the tree averages two child nodes; FD inverts an implicit BS stencil and projects onto the payoff (PSOR); LSM regresses realised discounted cash flows on basis functions to estimate continuation; and COS expands the value at each step in a cosine series whose coefficients $V_k(t_n)$ are recursed backward, split at the early-exercise point x_n^* (found by Newton from $c = \Lambda$) into a closed-form payoff piece G_k (the Lecture-6 χ_k, ψ_k) and a continuation piece C_k computed by FFT in $O(NK \log K)$. Finally, since all of these price a Bermudan, Richardson extrapolation on a few monitoring counts recovers the American limit.